

Introduction

The Library database management system will be a robust and easy to use tool to keep track of the transactions and inventory of books for both library customers and the staff at a library. The database system will keep track of inventory, allow users to sign in and manage their account, check out and return books, as well as pay fees. The staff can see and manage users accounts, check books in, add books, and miscellaneous administrative tasks that may need to be used to manage the traffic and inventory of books.

Relational Schema Mapping

- Clients
 - ID: INT; PK (Primary Key)
 - FirstName: VARCHAR
 - LastName: VARCHAR
 - PhoneNum: INT(10)
 - EmailAddress: VARCHAR
 - AccountStatus: ENUM("Good", "Limited", "Suspended")
 - MemberType: ENUM("General", "Student", "Senior")
 - BorrowRestrict: INT, FD: relies on MemberType and/or AccountStatus
- Transactions
 - ID: INT; PK
 - BorrowDate: DATE
 - ReturnDate: DATE
 - DueDate: DATE
 - Renewals: INT
 - FeeStatus: ENUM("NoFee", "Unpaid", "Paid")
 - ClientID: INT; FK(Clients.ID)
 - ItemID: INT; FK(Materials.ID)
- Reservations
 - ID: INT; PK
 - ReserveTime: DATETIME
 - ClientID: INT; FK(Clients.ID)
 - ItemID: INT; FK(Materials.ID)
- Materials
 - ID: INT; PK
 - AvailStatus: ENUM("Available", "Reserved", "CheckedOut")
- Books
 - ItemID: INT; PK; FK(Materials.ID)
 - Title: VARCHAR
 - Author: VARCHAR
 - Genre: VARCHAR
 - ISBN: INT(13)
 - PubDate: DATE
 - Rare: BOOL

- Digital Media
 - ItemID: INT; PK; FK(Materials.ID)
 - Title: VARCHAR
 - Creator: VARCHAR
 - Genre: VARCHAR
 - ISBN: INT(13)
 - PubDate: DATE
- Magazines
 - ItemID: INT; PK; FK(Materials.ID)
 - Title: VARCHAR
 - IssueNo: INT
 - PubDate: DATE
 - Rare: BOOL